Lecture Three

1 The System of Linear Equations

1.1 Basic Concepts

Definition 1.1.1 *A* ***linear equation*** *in the variables* $x_1, x_2, \dots, x_n$ *is an equation that can be written in the form*

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b \tag{1.1.1}$$

Where b *and the coefficients are real or complex numbers.*

For example, the equations

$$4x_1 - 5x_2 + 2 = x_1 \quad \text{and} \quad x_2 = 2\sqrt{6} - 2x_1 + x_3$$

are both linear because they can be rearranged algebraically as an equation 1.1.1

$$3x_1 - 5x_2 = -2 \quad \text{and} \quad 2x_1 + x_2 - x_3 = 2\sqrt{6}$$

But the equations

$$4x_1 - 5x_2 = x_1x_2 \quad \text{and} \quad x_2 = 2\sqrt{x_1} - 6$$

are not linear because of the presence of x_1x_2 in the first equation and $\sqrt{x_1}$ in the second one.

Definition 1.1.2 ***System of Linear Equations:*** *A system of* m *linear equations in* n *variables is a set of* m *equations each of which is linear in the same* n *variables. And it is in the form*

$$\begin{array}{c} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \cdots + a_{2n}x_n = b_2 \\ a_{31}x_1 + a_{32}x_x + a_{33}x_3 + \cdots + a_{3n}x_n = b_3 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \cdots + a_{mn}x_n = b_n \end{array} \tag{1.1.2}$$

Definition 1.1.3 ***Solution of a System of Linear Equations:*** *A solution of the system is a list* $(s_1, s_2, \dots, s_n)$ *of numbers that makes each equations a true statement when the values* $s_1, s_2, \dots, s_n$ *are substituted for* $x_1, x_2, \dots, x_n$ *respectively.*

Note:

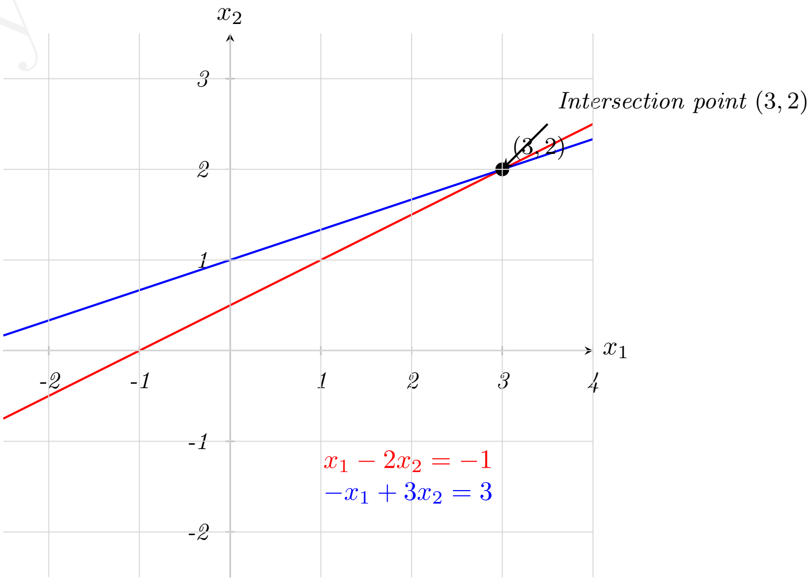
1. The set of all possible solutions is called the solution set of the linear system.
2. Two linear systems are called **equivalent** if they have the same solution set.

Finding the solution set of a system of two linear equations in two variables is easy because it amounts to finding the intersection of two lines.

Example 1.1.1 *For the system*

$$\begin{array}{rcrcr} x_1 & - & 2x_2 = & -1 \\ -x_1 & + & 3x_2 = & 3 \end{array}$$

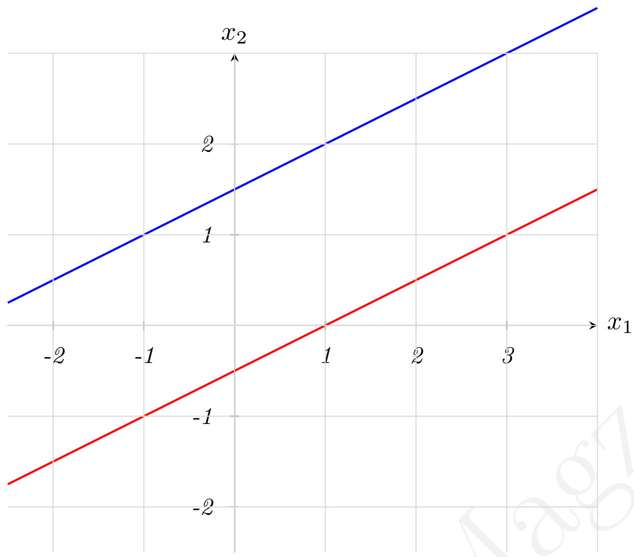
The graph of theses equations are lines. A pair of numbers (x_1, x_2) *satisfies both equations in the system if and only if the point* (x_1, x_2) *lies on both lines. In this system the solution is the single point* $(3, 2)$ *as shown below:*



Of course, two lines need not intersect in a single point ; they could be parallel, or they could be coincide and hence intersect at every point on the line. The following example shows this.

Example 1.1.2 *(a) The system*

$$\begin{array}{rcrcr} x_1 & - & 2x_2 & = & -1 \\ -x_1 & + & 2x_2 & = & 3 \end{array}$$

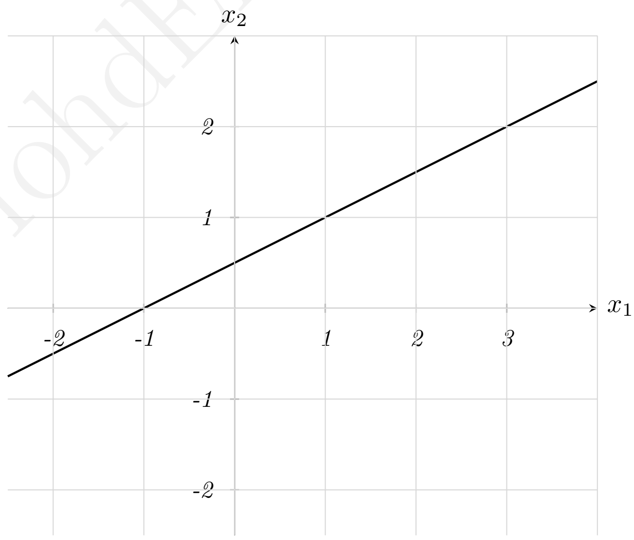


$l_1 : x_1 - 2x_2 = 1$
$l_2 : x_1 - 2x_2 = -3$
Parallel lines – no intersection

(b) The system

$$\begin{array}{rcrcr} x_1 & - & 2x_2 & = & -1 \\ -x_1 & + & 2x_2 & = & 1 \end{array}$$

has infinitely many solutions.

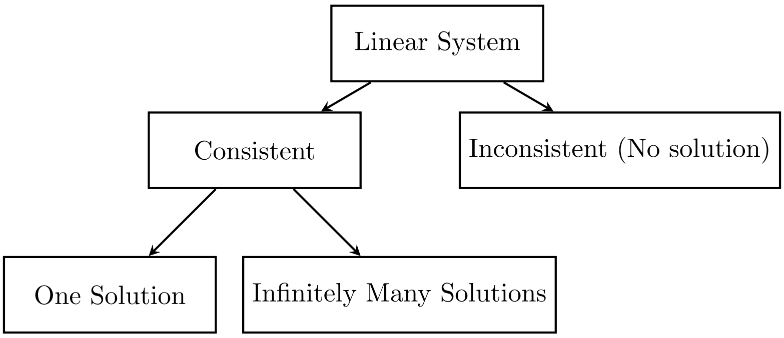


$x_1 - 2x_2 = -1$
Infinitely many solutions

A system of linear equation has

1. No solution, or
2. Exactly one solution, or
3. Infinitely many solutions.

A system of linear equations is said to be **consistent** if it has either one solution of infinitely many solutions; a system is **inconsistent** if it has no solution.



1.2 Matrix Notation

Given the system in equation 1.1.2, then the matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

is called **the coefficients matrix** or (matrix of coefficients) of the system 1.1.2. And the matrix

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} & b_2 \\ a_{31} & a_{32} & a_{33} & \dots & a_{3n} & b_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} & b_n \end{bmatrix}$$

is called **the augmented matrix** of the system.

Example 1.2.3 *For the system*

$$\begin{array}{rcrcrcr} x_1 & - & 2x_2 & + & x_3 & = & 0 \\ & & 2x_2 & - & 8x_3 & = & 8 \\ 5x_1 & & & - & 5x_3 & = & 10 \end{array}$$

The coefficients matrix is

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{bmatrix}$$

And the augmented matrix is

$$\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{bmatrix}$$

The augmented matrix of a system consists of the coefficient matrix with an added column containing the constants from the right side of the equations.

Example 1.2.4 *Determine if the following system is consistent.*

$$\begin{array}{rcrcrcr} & & 2x_2 & - & 4x_3 & = & 8 \\ 2x_1 & - & 3x_2 & + & 2x_3 & = & 1 \\ 4x_1 & - & 8x_2 & + & 12x_3 & = & 1 \end{array} \tag{1.2.1}$$

Sol

The augmented matrix for this system is

$$\begin{bmatrix} 0 & 1 & -4 & 8 \\ 2 & -3 & 2 & 1 \\ 4 & -8 & 12 & 1 \end{bmatrix}$$

To obtain an x_1 in the first equation, interchange rows 1 and 2.

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 4 & -8 & 12 & 1 \end{bmatrix}$$

To eliminate the $4x_1$ term in the third equation, add -2 times row 1 to row 3.

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & -2 & 8 & -1 \end{bmatrix}$$

Next, use the x_2 term in the second equation to eliminate the $-2x_2$ term from the third equation. Add 2 times row 2 to row 3.

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & 15 \end{bmatrix}$$

Now by going back to equation notation

$$\begin{array}{rcrcrcr} 2x_1 & - & 3x_2 & + & 2x_3 & = & 1 \\ & & x_2 & - & 4x_3 & = & 8 \\ & & & & 0 & = & 15 \end{array} \tag{1.2.2}$$

The equation $0 = 15$ is a short form of $0x_1 + 0x_2 + 0x_3 = 15$, and there are no values x_1, x_2, x_3 that satisfy this equation. Since 1.2.1 and 1.2.2 have the same solution set, the original system is inconsistent (has no solution).

Exercise 1.2.1 *Answer the following*

1. *The augmented matrix of a linear system has been transformed by row operations into the form below. Determine if the system is consistent.*

$$\begin{bmatrix} 1 & 5 & 2 & -6 \\ 0 & 4 & -7 & 2 \\ 0 & 0 & 5 & 0 \end{bmatrix}$$

2. *Is $(3, 4, -2)$ a solution of the following system*

$$\begin{array}{rcrcrcr} 5x_1 & - & x_2 & + & 2x_3 & = & 7 \\ -2x_1 & + & 6x_2 & + & 9x_3 & = & 0 \\ -7x_1 & + & 5x_2 & - & 3x_3 & = & -7 \end{array}$$

3. *For what values of h and k is the following system is consistent.*

$$\begin{array}{rcrcr} 2x_1 & - & x_2 & = & h \\ -6x_1 & + & 3x_2 & = & k \end{array}$$

1.3 Row Reduction and Echelon Form

Definition 1.3.4 *A rectangular matrix is in **echelon form** (or row echelon form) if it has the following properties.*

1. *All non zero rows are above any rows of all zeros.*
2. *Each leading entry of a row is in a column to the right of the leading entry of row above it.*
3. *All entries in a column below a leading entry are zeros.*

   *If a matrix in echelon form satisfies the following additional conditions, then it is in **reduced echelon form** or (reduced row echelon form):*
4. *The leading entry in each non zero row is 1.*
5. *Each leading 1 is the only non zero entry in its column.*

For example, the matrices

$$\begin{bmatrix} 2 & -3 & 2 & 1 \\ 0 & 1 & -4 & 8 \\ 0 & 0 & 0 & \frac{5}{2} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

are in echelon form. In fact the second matrix is in reduced echelon form.

Note: Any non zero matrix may be row reduced in echelon form from using different sequences of row operations. However, the row reduced echelon form one obtains from a matrix is unique.

Definition 1.3.5 *A **pivot position** in a matrix A is a location in A that corresponds to a leading 1 in the reduced echelon form of A. A **pivot column** is a column of A that contain a pivot position.*

Example 1.3.5 *Row reduce the matrix A below to echelon form and locate the pivot columns*

$$\begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & 1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$$

Sol

Begin with interchanging rows 1 and 4 to obtain

$$\begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & 1 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix}$$

Then create zero below a_{11}

$$\xrightarrow[R_3 \to R_3 + 2R_1]{R_2 \to R_2 + R_1} \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 5 & 10 & -15 & -15 \\ 0 & -3 & -6 & 4 & 9 \end{bmatrix} \xrightarrow[R_4 \to \frac{3}{2}R_2 + R_4]{R_3 \to -\frac{5}{2}R_2 + R_3} \begin{bmatrix} 1 & 4 & 5 & -9 & -7 \\ 0 & 2 & 4 & -6 & -6 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 & 0 \end{bmatrix}$$

Interchange row 3 and 4, we obtain

$$\begin{bmatrix} \boxed{1} & 4 & 5 & -9 & -7 \\ 0 & \boxed{2} & 4 & -6 & -6 \\ 0 & 0 & 0 & \boxed{-5} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The pivot positions a_{11}, a_{22}, a_{34}, and the pivot columns are column 1,2 and 4. Therefore $0, -2$ and 3 are the pivot positions in the given matrix.

1.4 Solution of Linear System

Example 1.4.6 *Suppose that the augmented matrix of a linear system has been changed into the equivalent reduced echelon form*

$$\begin{bmatrix} 1 & 0 & -5 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are three variables because the augmented matrix has four columns. The associated system of equations is:

$$\begin{array}{ccccccc} x_1 & & & - & 5x_3 & = & 1 \\ & & x_2 & + & x_3 & = & 4 \\ & & & & 0 & = & 0 \end{array}$$

The variable x_1 and x_2 corresponding to picot columns in the matrix are called (Basic Variables). The other variable, x_3 is called a (free variable).

$$\begin{cases} x_1 = 1 + 5x_3 \\ x_2 = 4 - x_3 \\ x_3 \text{ is free} \end{cases}$$

The statement x_3 is free means that you are free to choose any value for x_3, for instance, when $x_3 = 0 \Rightarrow x_1 = 1 + 5(0) = 1, x_2 = 4 - 0 = 4$. Therefore $(1, 4, 0)$ is a solution of the system.

Example 1.4.7 *Find the general solution of the linear system whose augmented matrix has been reduced to*

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

Sol

$$\begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & -1 & 3 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix} \xrightarrow{R_3 \to R_2 + R_3} \begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 2 & -8 & 0 & 10 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix} \xrightarrow{R_2 \to \frac{1}{2} R_2} \begin{bmatrix} 1 & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & 1 & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 7 \end{bmatrix}$$

$$\xrightarrow{R_1 \to R_1 + (-2) R_2} \begin{bmatrix} \boxed{1} & 6 & 2 & -5 & -2 & -4 \\ 0 & 0 & \boxed{1} & -4 & 0 & 5 \\ 0 & 0 & 0 & 0 & \boxed{1} & 7 \end{bmatrix}$$

There are five variables because the augmented matrix has six columns. The pivot columns of the matrix are 1,3 and 5. So the basic variables are x_1, x_3 and x_5. The remaining variables x_2 and x_4 must be free. Now, the associated system of equations is

$$\begin{aligned} x_1 + 6x_2 + 3x_4 &= 0 \\ x_3 - 4x_4 &= 5 \\ x_5 &= 7 \end{aligned}$$

Solve for the basic variables to obtain the general solutions.

$$\begin{cases} x_1 = -x_2 - 3x_4 \\ x_2 \text{ is free} \\ x_3 = 5 + 4x_4 \\ x_4 \text{ is free} \\ x_5 = 7 \end{cases}$$

Theorem 1.4.1 ***Existence and Uniqueness Theorem*** *A linear system is consistent if and only if the right-most column of the augmented matrix is not a pivot column. That is, if and only if an echelon form of the augmented matrix has no row of the form $[0 \;\; 0 \;\; 0 \; \ldots \; b]$, with $b \neq 0$.*

If a linear system is consistent, then the solution set contains either

(a) a unique solution when there are no free variables. or

(b) infinitely many solutions when there is at least one free variable.

The following procedure outlines how to find and describe all solutions of a linear system.

1. Write the augmented matrix of the system.
2. Use the row reduction algorithm to obtain an equivalent augmented matrix in echelon form. Decide whether the system is consistent . If there is no solution, stop; otherwise go to the next step.
3. Continue row reduction to obtain the reduced echelon form.
4. Write the system of equations corresponding to the matrix obtained in step 3.
5. Rewrite each non zero equation from step 4 so that its one basic variable is expressed in terms of any free variables appearing in the equation.

Exercise 1.4.2 *Answer the following*

1. *Find the general solution of the linear system whose augmented matrix is*

$$\begin{bmatrix} 1 & -3 & -5 & 0 \\ 0 & 1 & -1 & -1 \end{bmatrix}$$

2. *Find the general solution of the system*

$$\begin{array}{rcrcrcrcr} x_1 & - & 2x_2 & - & x_3 & + & 3x_4 & = & 0 \\ -2x_1 & + & 4x_2 & + & 5x_3 & - & 5x_4 & = & 3 \\ 3x_1 & - & 6x_2 & - & 6x_3 & + & 8x_4 & = & 2 \end{array}$$

3. *Choose h and k such that the system has:*

   (a) no solution

   (b) a unique solution

   (c) many solutions

   Give separate answers for each part.

   (i) $\begin{aligned} x_1 + hx_2 &= 1 \\ 4x_1 + 8x_2 &= k \end{aligned}$

   (ii) $\begin{aligned} x_1 + 3x_2 &= 2 \\ 3x_1 + hx_2 &= k \end{aligned}$

1.5 Cramer's Rule

It uses determinants to solve a system of n linear equations in n variables. This rule applies only to systems with unique solutions. To see how Cramer's rule works, look at the system:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{aligned}$$

it has the solution $x_1 = \dfrac{b_1a_{22} - b_2a_{12}}{a_{11}a_{22} - a_{21}a_{12}}$ and $x_2 = \dfrac{b_2a_{11} - b_1a_{21}}{a_{11}a_{22} - a_{21}a_{12}}$, when $a_{11}a_{22} - a_{21}a_{12} \neq 0$. A determinant can represent each numerator and denominator in this solution as shown below.

$$x_1 = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}} \quad , \quad x_2 = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}} \qquad , a_{11}a_{22} - a_{21}a_{12} \neq 0$$

Note

1. The denominator for x_1 and x_2 is simply the determinant of the coefficient matrix A of the original system.
2. The numerator for x_1 and x_2 are formed by using the column of constants as replacements of the coefficients of x_1 and x_2 in $|A|$. These two determinants are denoted by $|A_1|$ and $|A_2|$ as shown below

$$|A_1| = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix} \quad \text{and} \quad |A_2| = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}$$

We have $x_1 = \dfrac{|A_1|}{|A|}$, $x_2 = \dfrac{|A_2|}{|A|}$. This determinant form of the solution is called **Cramer's Rule**.

Example 1.5.8 *Solve the following system*

$$\begin{aligned} 4x_1 - 2x_2 = 10 \\ 3x_1 - 5x_2 = 11 \end{aligned}$$

Sol

Firstly, we find the determinant of the coefficient matrix.

$$|A| = \begin{vmatrix} 4 & -2 \\ 3 & -5 \end{vmatrix} = -14 \ \neq 0$$

So the system has a unique solution, and applying Cramer's rule produces:

$$x_1 = \frac{|A_1|}{|A|} = \frac{\begin{vmatrix} 10 & -2 \\ 11 & -5 \end{vmatrix}}{-14} = \frac{-28}{-14} = 2 \quad \textit{and} \quad x_2 = \frac{|A_2|}{|A|} = \frac{\begin{vmatrix} 4 & 10 \\ 3 & 11 \end{vmatrix}}{-14} = \frac{14}{-14} = -1$$

Therefore the solution is $x_1 = 2$ and $x_2 = -1$.

Cramer's Rule generalizes to systems of $n-$ linear equations in n variables. The value of each variable is the quotient of two determinants. The denominator is the determinant of the coefficient matrix, and the numerator is the determinant of the matrix formed by replacing the column corresponding to the variable being solved for with the column representing the constants. For example, the value of x_3 in the system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{aligned} \quad \text{is} \quad x_3 = \frac{\begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}}$$

Theorem 1.5.2 *If a system of n linear equation in $n-$ variables has a coefficient matrix A with a non zero determinant $|A|$, then the solution of the system is*

$$x_1 = \frac{\det(A_1)}{\det(A)}, \ x_2 = \frac{\det(A_2)}{\det(A)}, \ \ldots, x_n = \frac{\det(A_n)}{\det(A)}$$

Where the ith column of A_i is the column of constants in the system of equations.

Example 1.5.9 *Solve the following system of linear equations:*

$$\begin{aligned} -x + 2y - 3z &= 1 \\ 2x + z &= 0 \\ 3x - 4y + 4z &= 2 \end{aligned}$$

Sol

The determinant of the coefficient matrix is $|A| = \begin{vmatrix} -1 & 2 & -3 \\ 2 & 0 & 1 \\ 3 & -4 & 4 \end{vmatrix} = 10$. *So solution is unique.*

$$|A_1| = \begin{vmatrix} 1 & 2 & -3 \\ 0 & 0 & 1 \\ 2 & -4 & 4 \end{vmatrix} = 8 \ , \ |A_2| = \begin{vmatrix} -1 & 1 & -3 \\ 2 & 0 & 1 \\ 3 & 2 & 4 \end{vmatrix} = -15 \ , \ |A_3| = \begin{vmatrix} -1 & 2 & 1 \\ 2 & 0 & 0 \\ 3 & -4 & 2 \end{vmatrix} = -16$$

Now

$$x = \frac{8}{10} = \frac{4}{5}, \ y = \frac{-15}{10} = \frac{-3}{2}, \ z = \frac{-8}{5}$$

1.6 Applications of Determinant

Determinants have many applications in analytic geometry, one is finding the area of a triangle in the $xy-$plane.

Definition 1.6.6 *The* ***area of the triangle*** *with vertices* $(x_1, y_1), (x_2, y_2)$ *and* (x_3, y_3) *is:*

$$Area \ = \pm\frac{1}{2} \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix}$$

Where the sign $(\pm)$ *is chosen to give a positive area.*

Example 1.6.10 *Find the area of the triangle whose vertices are* $(1, 1), (2, 2)$ *and* $(4, 3)$.

Sol

Simply evaluate the determinant

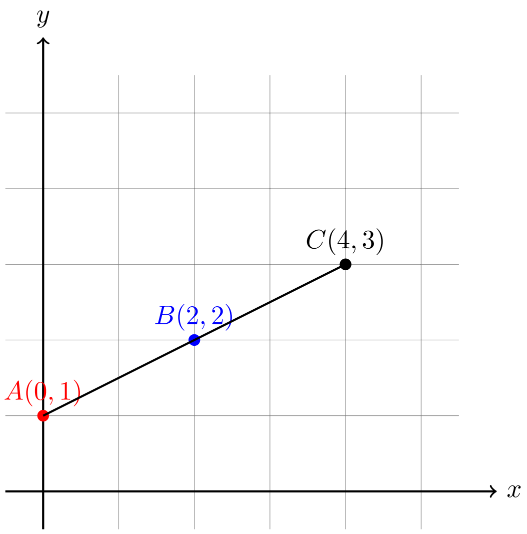
$$\frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 4 & 3 & 1 \end{vmatrix} = -\frac{1}{2}$$

. *And conclude that the area of the triangle is* $\frac{1}{2}$ *square unit.*

If the three points in example 1.6.10 had been on the same line what would have happened when you applied the area formula?

Consider, for example the three collinear points $(0, 1), (2, 2)$ and $(4, 3)$. The determinant that yields the area of the triangle that has these three points as vertices is

$$\frac{1}{2}\begin{vmatrix} 0 & 1 & 1 \\ 2 & 2 & 1 \\ 4 & 3 & 1 \end{vmatrix} = 0$$



Theorem 1.6.3 ***Test for Collinear Points in the xy plane:*** *Three points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) are collinear if and only if*

$$\det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix} = 0$$

Theorem 1.6.4 ***Two points Form of an equation of a Line:*** *An equation of the line passing through the distinct points (x_1, y_1) and (x_2, y_2) is*

$$\det \begin{bmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{bmatrix} = 0$$

Example 1.6.11 *Find an equation of line passing through the points $(2, 4)$ and $(-1, 3)$*

Sol

Let $(x_1, y_1) = (2, 4)$ and $(x_2, y_2) = (-1, 3)$. Applying the determinant formula for an equation of a line produces:

$$\begin{vmatrix} x & y & 1 \\ 2 & 4 & 1 \\ -1 & 3 & 1 \end{vmatrix} = 0$$

Expand by cofactors in the first row, we get

$$x \begin{vmatrix} 4 & 1 \\ 3 & 1 \end{vmatrix} - y \begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} + 1 \begin{vmatrix} 2 & 4 \\ -1 & 3 \end{vmatrix} = 0 \Rightarrow x - 3y + 10 = 0$$